Session 24
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Module 4, Week 11
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Regression and Classification

Using

Sklearn (A tool kit/ML-package in python)

https://colab.research.google.com/drive/1DxRG2Q9mSYhRiIZS7qhP8GGlwVe6j6Wg?usp=sharing

Objectives

- Understand the concept of Multiple Linear Regression (MLR).
- Differentiate between Simple and Multiple Linear Regression.
- Build a Multiple Linear Regression model using Scikit-learn.
- Train and evaluate a regression model with multiple predictors.
- Interpret regression coefficients and model performance.

Why Multiple Linear Regression?

- Real world is little more complex
- One output variable can depend on many input variables.
- For eg.,

  House price can depend on *number of bed rooms, age of the house, size of the house, number of amenities around the society, location of the area, etc.*

  Student Performance in coming exams may depend on Study Hours, Attendance in the course, Previous GPA, etc.

  Salary Prediction for a new joinee may depend on Experience, Education, Leadership score, other Skills.

Multiple Linear Regression allows us to model multiple predictors/input variables

Simple Linear Regression (SLR) Vs Multiple Linear Regression (MLR)

- If multiple predictors are there, Multiple Linear Regression may **assure more accuracy** in prediction.
- Simple linear regression **can be visualized**, while MLR is more than two dimensions.
- Simple linear regression is the **foundation to understand** the concept of regression algorithm, MLR is built on top of the idea of simple linear regression.
- More features/predictors assures more accurate system, hence MLR is preferable over SLR if we have more features.

Modeling multiple linear regression

$y = mx+c$ or $y = f(x)$

//y is a function of x with unknown parameters m and c, where m is the slope (coefficient) of the line, c is the intercept (value of y at x=0)

$y = w_0+w_1x_1+w_2x_2 + w_3x_3 + w_4x_4 + \ldots\ldots\ldots\ldots w_nx_n$ or $y = f(x0, x_1, x_2 \ldots\ldots\ldots x_n)$ //m and c are represented as weight (w) parameters in ML

where,

- y = *dependent variable*
- *x1,x2,....,,xn = independent variables/predictors*
- w_0 = *intercept*
- $w_1w_2w_{3...}w_n$ = *coefficients/weights*

Modeling multiple linear regression

Salary = 25000 + 4000*(Experience) + 3000 * (Education) + 500 * (CS)

w0 = 25000 (base salary without any x), w1 = 4000, w2 = 3000, w3 = 500

If input is:

Experience (x_1) = 5 years, Education (x_2) = 2 (Master's), Communication skill score out of 100 (x_3) = 30

Predicted Salary =25000+4000(5)+3000(2)+500(30) = 25000+20000+6000+15000 =66000

- Each feature contributes independently to the final prediction in proportion of the weights.
- The job of multiple linear regression algorithm is to learn optimal values of weights/coefficients

How to implement LR in Python?

sklearn ← Package

Linear_model ← Module

LinearRegression ← Having Functions for doing LR

```
from sklearn.linear_model import LinearRegression
```

- Sklearn Package (library) is The Scikit-learn machine learning package in python.
- Linear_model is a module inside sklearn containing linear machine learning models.
- LinearRegression is a class for creating Linear Regression model objects

Practice Example //Optional slide

[w1 = 7509.57061562, w2 = -785.30781169 , w3 = 104.500776]

W0 = 18600.103466114895

Predicted_salary (9th data point) = **w0+w1*9+w2*3+w3*48** = **88846**

Test data point = 9th point, with experience of 9 years and salary is 90000

Errors and their interpretation

Lower MAE → Better predictions

Lower MSE → Smaller prediction errors

R^2 closer to 1 → Better fit

Performance Evaluation

Three suitable metrics

1. Mean Absolute Error (MAE): Measures average absolute prediction error, lower is better.
2. *Mean Square Error (MSE): Measure average square prediction error, lower is better //suitable for finding optimal parameter values also

   Root Mean Squared Error (RMSE): Take root of the 2

3. R^2 Score: Measures goodness of fit between actual y and predicted y, higher is better, maximum value is 1.

Limitations of LR

- Will fail for data having complex relationship with y, other more sophisticated algorithms are the solution

- Outliers may spoil the model

Thank you!